

Luís Pedro Borges

Machine Learning | Information Retrieval | Neural Search

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Open to jobs and internships in Canada and Europe.

Canada: IEC Working Holiday approved for up to 2 years.

Open work permit activation on arrival; no employer sponsorship required during the permit period.

Profile

Machine learning and information retrieval researcher with a **Ph.D. in progress** at Instituto Superior Técnico and Carnegie Mellon University. Experience in neural retrieval, knowledge distillation and LLM re-ranking, including benchmark experiments at scale and search efficiency evaluation. Seeking Research Scientist, Data Scientist, Data Engineer, ML Engineer and Search Engineer roles and internships in Canada and Europe.

Technical Skills

Programming: Python, C++, Java, C.

Machine learning: PyTorch, TensorFlow, Keras, NumPy, SciPy, scikit-learn.

Search and NLP: learned sparse and dense retrieval, inverted indexes, PISA, passage re-ranking, knowledge distillation, LLM re-ranking, named entity recognition, fact verification.

Evaluation: MS MARCO, TREC Deep Learning and BEIR; ranking quality, recall and retrieval latency.

Research Experience

Early Stage Researcher

Mar 2025 - Present

INESC-ID, Lisbon, Portugal

- Research knowledge distillation methods for small language models in information retrieval.

Doctoral Research - Ph.D. in progress

Sep 2019 - Present

Instituto Superior Técnico / INESC-ID and Carnegie Mellon University

- Developed **KALE**, connecting learned sparse query and document representations with efficient inverted index search; evaluated on MS MARCO, TREC and BEIR.
- Conducted PyTorch distillation experiments using approximately **45 million existing synthetic query-passage judgments**, with training on a single NVIDIA A100 GPU.
- Developed and evaluated **EmberTiny**, a retriever with 14.5M parameters that retained **97% of the MRR@10 of its SimLM teacher with 110M parameters** on MS MARCOv1 (0.403 versus 0.415).
- Developed **SCALE** and evaluated it with **SPLADE.Lex**: **44% lower mean index search latency** than SPLADEv3 (60 versus 108 ms), with MRR@10 improving from 0.400 to 0.415 in MS MARCOv1 dissertation experiments.
- Developed retrieval and zero-shot LLM re-ranking for Tip-of-the-Tongue search across domains; published at SIGIR 2024, with public code and data.

Additional Experience

Teaching Assistant - Search Engines

Jan 2022 - May 2022

Carnegie Mellon University, Pittsburgh, United States

- Supported the Search Engines course under Professor Jamie Callan.

Junior Researcher

Mar 2019 - Sep 2019; Jul 2018 - Dec 2018

INESC-ID, Lisbon, Portugal

- Worked on string matching tasks in natural language processing across two research appointments.

Visiting Junior Researcher

Jan 2019 - Apr 2019

National Institute of Informatics, Tokyo, Japan

- Worked on the FEVER fact verification task.

Junior Researcher Intern

Jul 2018 - Sep 2018

Priberam, Lisbon, Portugal

- Worked on the OntoNotes 5.0 named entity recognition task.

Education

Ph.D. in Language and Information Technologies - in progress

2019 - Present

Instituto Superior Técnico and Carnegie Mellon University

Dual degree programme. Advisors: Bruno Martins (IST) and Jamie Callan (CMU).

Dissertation research: *Latent Vocabularies for Learned Sparse Retrieval*.

M.Sc. in Information Systems and Computer Engineering

2016 - 2018

Instituto Superior Técnico, University of Lisbon

Thesis: *Detecting Fake News Using Deep Neural Networks*.

B.Sc. in Information Systems and Computer Engineering

2013 - 2016

Instituto Superior Técnico, University of Lisbon

Selected Publications

Generalizable Tip-of-the-Tongue Retrieval with LLM Re-ranking

L. Borges, R. Jha, J. Callan and B. Martins. *SIGIR*, 2024.

Paper

KALE: Using a K-Sparse Projector for Lexical Expansion

L. Borges, B. Martins and J. Callan. *ICTIR*, 2023.

Paper

Assessing the Benefits of Model Ensembles in Neural Re-ranking for Passage Retrieval

L. Borges, B. Martins and J. Callan. *ECIR*, 2021.

Paper

Full publication list and open papers